



APPLICOM

AZURE MANAGEMENT API WITH JAVA

APPLICOM – SYSTEMS INTEGRATION AND BEYOND

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ABOUT APPLICOM



- They develop and maintain **software** for their clients
- They build complete **IT infrastructure systems**
- Their clients operate within the following industries:
healthcare, finance, insurance, oil & gas, industrial,
telecommunications, manufacturing, media and logistics

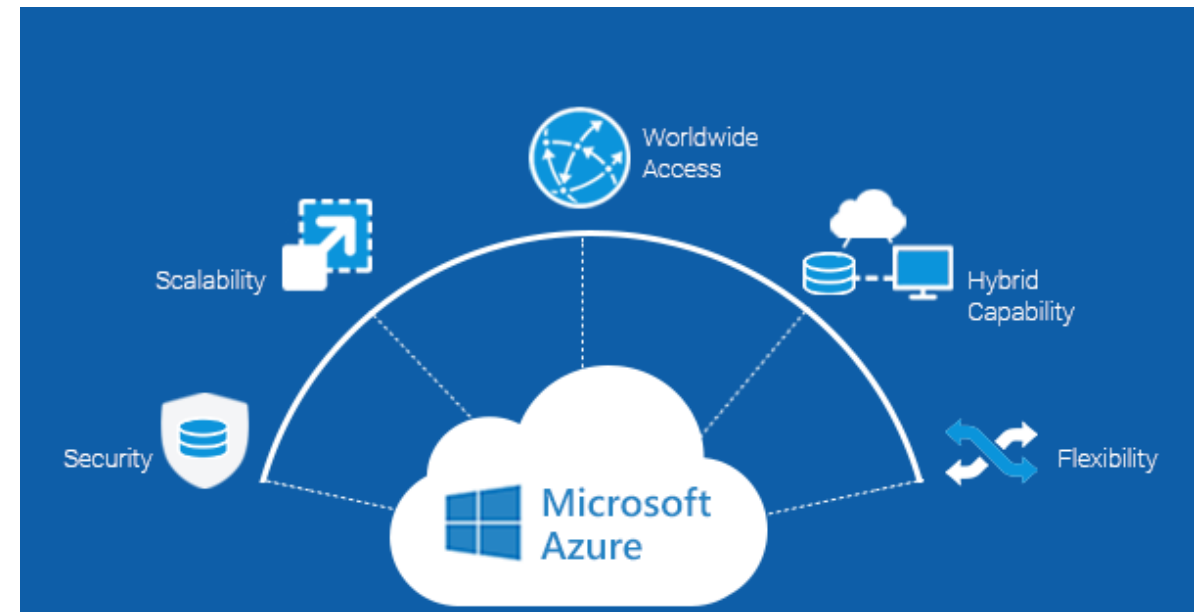


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WHAT IS AZURE?



- Public **cloud platform** with datacenters all over the world
- You still have to manage your spending when maintaining your VM, just like traditional datacenters
- Their **API** allows companies to optimize + automate datacenter tasks



MY PROJECT

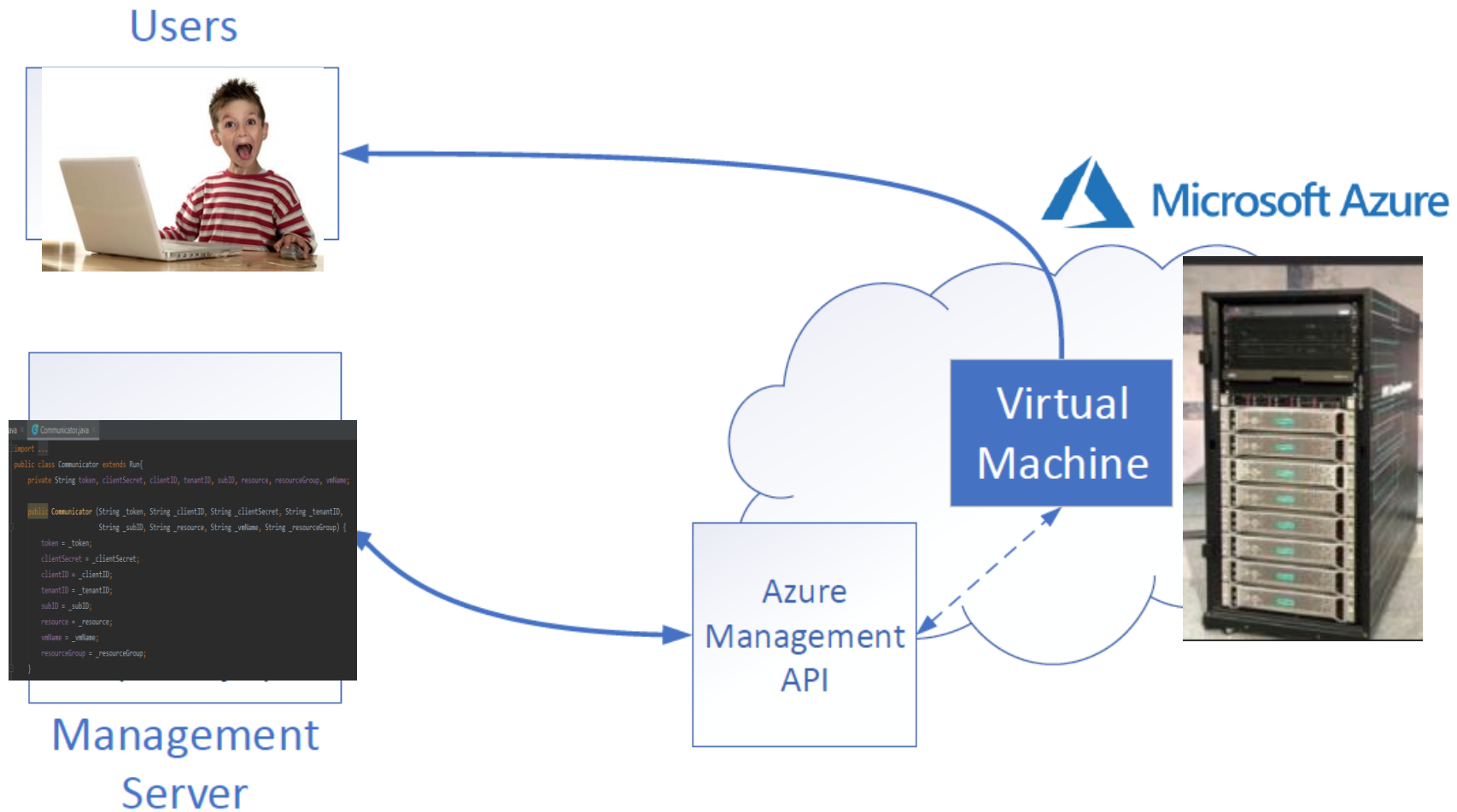
- I used Azure to rent out my own VM from their Dubai datacenter.
- My objective was to resize the Virtual Machine's processor so that it'd **shrink during the weekdays** and **increase during the weekends**.
 - This would, for this theoretical client, optimize spending by cutting back on unused resources.
 - To achieve this, I wrote code for API queries in **Java**, and used **Task Scheduler** to automate the process.
- This project balances **business** and **technical** skills that Applicom utilizes
 - Also deals with the same technology (e.g., VMs, API queries)

KEY COMPONENTS



- Azure Environment (CLI)
- A Virtual Machine (which I rented from Microsoft)
- Postman, an online tool used for testing the below API queries:
 - **1. Get Size**
 - **2. Get VM**
 - **3. Create a Resource Group**
 - **4. Resize the VM**
 - **5. Get Bearer Token**
- Java application with the below features
 - **1. Resize VM manually (according to user input)**
 - **2. Resize VM (according to “args”)**
- Windows Task Scheduler job
 - **1. Automate the resizing of the VM**

DIAGRAM OF THE PROCESS



GET

Azure REST 2021 (Keep) / Get Resource Groups

GET



https://management.azure.com/subscriptions/{{subscriptionId}}/resourcegroups?api-version=2020-09-01

Params ●

Authorization

Headers (6)

Body

Pre-request Script

Tests

Settings

Query Params

	KEY	VALUE
☒	api-version	2020-09-01
	Key	Value

POST

Azure REST 2021 (Keep) / getToken

POST

▼

https://login.microsoftonline.com/{{tenantId}}/oauth2/token

Params

Authorization

Headers (8)

Body ●

Pre-request Script

Tests

Settings

☐ none

☐ form-data

☒ x-www-form-urlencoded

☐ raw

☐ binary

☐ GraphQL

	KEY	VALUE
☒	grant_type	client_credentials
☒	client_id	296b1ec5-e26b-414c-b1cb-bbe9d3ecdcc1
☒	client_secret	1X4FSU-T-vO88se7kzizm~cK9XY1TDIaDz
☒	resource	https://management.azure.com/

PUT

Azure REST 2021 (Keep) / Resize VM



Save



...

PUT



https://management.azure.com/subscriptions/{{subscriptionId}}/resourceGroups/SHARMOOT_GROUP/providers/Microsoft.Compute/virtualMachines/Sharmoot?api-version=2021-04-01

Params ●

Authorization

Headers (8)

Body ●

Pre-request Script

Tests

Settings

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL **JSON** ▼

```
1 {  
2   "properties":{  
3     "hardwareProfile":{  
4       "vmSize": "Standard_B4ms"  
5     }  
6   },  
7   "location": "UAE-North"  
8 }
```

TASK SCHEDULER



- *Increase* virtual machine size to **B4ms** at the start of the weekend (Friday evening) **to ensure there are enough resources to support the additional user load.**
- *Decrease* virtual machine size to **B2ms** at the start of the week (Monday morning) **to reduce cost.**

 Resize to B2ms	Ready	At 4:30 AM every Monday of every week, starting 8/3/2021
 Resize to B4ms	Ready	At 7:30 PM every Friday of every week, starting 8/3/2021